

Thesis -> Paper

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Abstract

This thesis is being transformed into a paper

1 Introduction

1.1 Motivation and Research Context

The mean-variance framework introduced by Markowitz [1952] has shaped capital allocation for over seven decades, yet its core assumptions of stable correlations and reliable parameter estimates break down in exactly the conditions where diversification is meant to help. During market declines, correlations among S&P 500 constituents rise well above their upside levels, an asymmetry that erodes the diversification the framework is built to deliver precisely when protection matters most [Ang and Chen 2002].

2 Regimes

2.1 Changepoint Detection

Methodology and calculation 1:1 to the paper

2.2 HMM

2.3 Wasserstein

3 References

Nystup Peter, Madsen Hendrik, Lindström Erik, 2017. Dynamic portfolio optimization across hidden market regimes.

Used for:

- Transaction costs 10bps
- Hidden Markov Model